

# Sales Data Analysis Using DAX and Data Visualisation

## Objective:

This project focuses on analyzing sales data using Power BI, DAX measures, and interactive visualizations. The dataset was cleaned, transformed, and modeled to generate meaningful business insights. Various DAX calculations such as Total Sales, Average Sales, Profit Analysis, Order Count, and Year-to-Date (YTD) Sales were created to evaluate business performance. Interactive dashboards and visual reports were developed to track sales trends, customer behavior, regional performance, and key performance indicators (KPIs), enabling data-driven decision-making.

## Dataset Description:

The dataset contains the following columns.

### Column description:

| Column Name         | Description                        |
|---------------------|------------------------------------|
| Order Id            | Order ID of the Product            |
| Order Date          | Order Date of the Product          |
| Customer Name       | Customer Name of the Product       |
| State               | State Name of the Customer         |
| City                | City Name of the Customer          |
| Amount              | Price of the Product               |
| Profit              | Profit of the Product              |
| Quantity            | Quantity of the Product            |
| Category            | Category of the Product            |
| Sub-Category        | Sub-Category of the Product        |
| Month of Order Date | Order Date of the Sales            |
| Target              | Target for the Sales               |
| Profit Margin       | Newly Calculated the Profit Margin |
| Date                | Newly created Date of the table    |

|              |                                        |
|--------------|----------------------------------------|
| Year         | Newly created Year of the Date         |
| Month Number | Newly created Month Number of the Date |
| Month        | Newly created Month name of the Date   |
| YearMonth    | Newly created Year Month format        |

## Formulas Used:

### Calculated Columns

#### Create a Calculated Column for 'Category Type':

Category Type = 'Order Details (2)'[Category]&"-"&'Order Details (2)'[Sub-Category]

#### Calculate Revenue per Order in Order Details Table:

Revenue = 'Order Details (2)'[Amount]\*'Order Details (2)'[Quantity]

#### Create a Calculated Column to Categorize Sales:

Sales Category = IF('Order Details (2)'[Amount]>AVERAGE('Order Details (2)'[Amount]),"Above Average","Below Average")

### Calculated Measures

#### Average Profit in Delhi:

Average Profit in Delhi = CALCULATE(AVERAGE('Order Details (2)'[Profit]),'List of Orders (2)'[City]="Delhi")

#### Order Count:

Order Count = COUNT('Order Details (2)'[Order ID])

### Year-to-Date (YTD) Sale:

Year to Date Sales = `TOTALYTD(SUM('Orders Data'[Sales Revenue]), 'Date'[Date])`

| Year  | Year to Date Sales |
|-------|--------------------|
| 2018  | ₹ 13,10,271        |
| 4     | ₹ 1,52,827         |
| 5     | ₹ 2,91,594         |
| 6     | ₹ 4,01,609         |
| 7     | ₹ 4,54,753         |
| 8     | ₹ 6,06,267         |
| 9     | ₹ 7,38,984         |
| 10    | ₹ 9,15,647         |
| 11    | ₹ 11,45,432        |
| 12    | ₹ 13,10,271        |
| 2019  | ₹ 8,36,599         |
| 1     | ₹ 3,37,229         |
| 2     | ₹ 5,14,849         |
| 3     | ₹ 8,36,599         |
| Total | ₹ 8,36,599         |

## Calendar Date Table Creation

Date =

```
ADDCOLUMNS(  
    CALENDAR(DATE(2018,4,1), DATE(2019,3,31)),  
    "Year", YEAR([Date]),  
    "Month Number", MONTH([Date]),  
    "Month", FORMAT([Date], "MMM"),  
    "YearMonth", FORMAT([Date], "YYYY-MM")  
)
```

| Structure     | manage relationships | view measure | search measure | column table | table     |
|---------------|----------------------|--------------|----------------|--------------|-----------|
| Relationships |                      |              |                | Calculations | Calendars |

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1 Date =

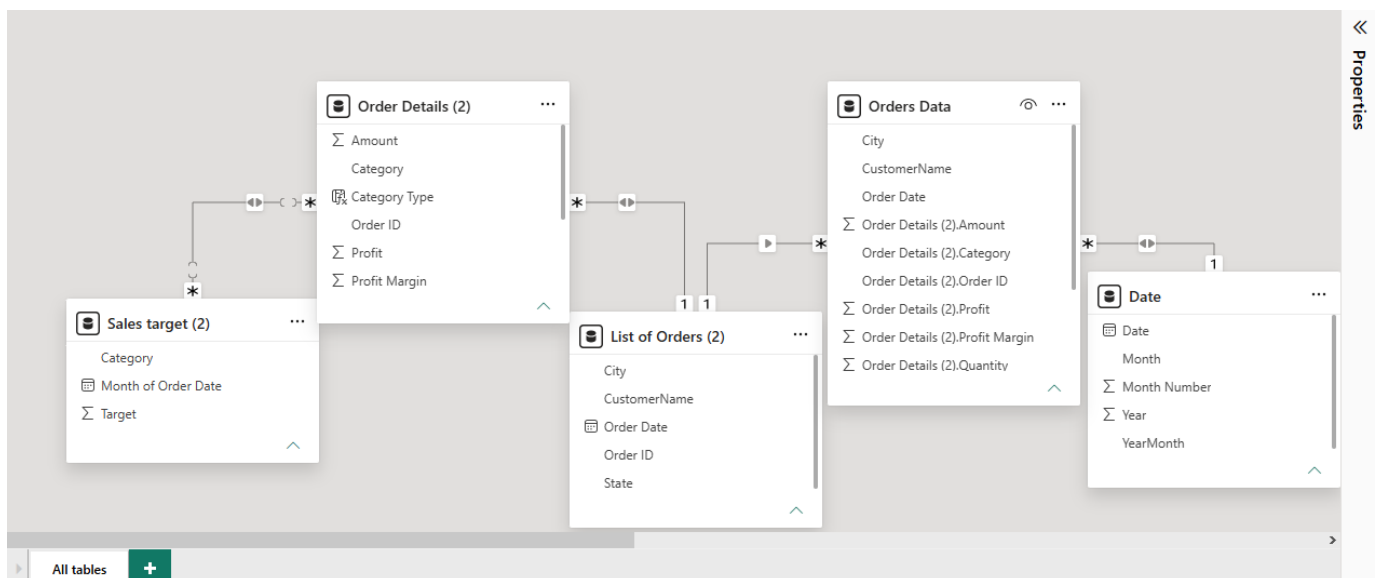
| Date                | Year | Month Number | Month | YearMonth |
|---------------------|------|--------------|-------|-----------|
| 01-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 02-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 03-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 04-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 05-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 06-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 07-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 08-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 09-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 10-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 11-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 12-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 13-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 14-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 15-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 16-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 17-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 18-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 19-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 20-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 21-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 22-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |
| 23-04-2018 00:00:00 | 2018 | 4            | Apr   | 2018-04   |

Table: Date (365 rows)

## Conclusion:

The project demonstrates best practices in DAX calculations and data visualization that support effective business intelligence and decision-making.

## Data Modelling:



Dashboard Layout:

